Firewalla Gold SE to cloud observability: a complete pipeline guide

The Firewalla Gold SE can export rich network telemetry to cloud-hosted Grafana and log platforms, but requires hands-on configuration since it lacks native monitoring integrations. **The optimal cost-free architecture pairs Telegraf + Fluent Bit on the Firewalla with Grafana Cloud's free tier**, which provides 10,000 metric series and 50 GB/month of log ingestion with 14-day retention (X) — more than sufficient for a single home firewall. The device's Debian-based Linux OS, SSH access, Docker support, and open-source codebase make this achievable, though the overlay filesystem and 4 GB RAM constraint require careful planning. No official Grafana or SIEM integration exists, and the Firewalla team has acknowledged this as a deprioritized feature request for over five years. (Firewalla) (Firewalla)

What the Gold SE actually exposes under the hood

The Firewalla Gold SE runs a quad-core ARM64 processor with **4 GB DDR4 RAM** and **32 GB eMMC storage**. Its software stack is open source (AGPL-3.0, at (github.com/firewalla/firewalla)) (Firewalla) and built on Node.js, Redis, and — critically for monitoring — **Zeek** (formerly Bro) for deep packet inspection. (Firewalla) This means the device generates structured network flow data far richer than basic syslog.

The most valuable data lives in **Zeek log files** at two locations. Active logs stream to (/bspool/manager/) in real-time plaintext, including (conn.log) (every TCP/UDP/ICMP connection with source, destination, port, protocol, duration, and byte counts), (dns.log) (all DNS queries and responses), (ssl.log) (TLS handshakes with SNI and certificate details), (http.log) (HTTP request metadata), and (files.log) (observed file transfers). Archived rotated logs land in (/log/blog/) as gzip-compressed JSON. (github) (Firewalla) Firewalla application logs reside under (/alog/firewalla/) (FireMain.log, FireAPI.log, FireMon.log, FireRouter.log) and (/alog/) contains ACL event logs ((acl-alarm.log), (acl-audit.log), (dnsmasq-acl.log)). (github) Standard system syslog lives at (/var/log/syslog).

Redis serves as the primary internal data store on localhost:6379 (thisDaveJ) (no authentication), holding device inventories keyed by MAC address ((host:mac:*)), aggregated flow data ((flow:*), (monitor:flow:*)), security alarms ((alarm:*)), firewall policies ((policy:*)), and threat intelligence ((intel:*)). These key patterns are reverse-engineered from the open source code and not officially documented.

The device has significant export limitations. **No native syslog forwarding, SNMP, NetFlow/sFlow, or MQTT** is built in. The Firewalla MSP API (cloud-based REST, requiring a ~$39/year subscription (Firewalla) at (docs.firewalla.net)) provides programmatic access to flows, alarms, devices, and rules (Firewalla) with endpoints like (POST /v2/flows/query) supporting time-range and device filters. (Firewalla) Without MSP, all data extraction requires SSH-level access. (Firewalla) (Firewalla)

The overlay filesystem is the biggest operational gotcha. Firewalla mounts a read-only base OS at (/media/root-ro) with a tmpfs writable overlay (Firewalla) — **every (apt install) is lost on reboot**. Firmware auto-updates occur roughly every two months and reset the overlay. The designated persistence path is (/home/pi/.firewalla/config/post_main.d/), where shell scripts execute automatically after every reboot and update. (Firewalla) (GitHub) Custom cron jobs survive via (/home/pi/.firewalla/config/user_crontab). Docker containers with restart policies also persist.

Building the metrics pipeline with Telegraf and Grafana Cloud

Grafana Cloud's free tier is the clear winner for a home user's metrics backend. It offers **10,000 active Prometheus series** with 14-day retention, 50 GB/month of Loki log ingestion, X full Grafana visualization with 3 users, X and built-in alerting — no credit card required. Grafana A typical Firewalla monitoring setup consumes roughly **150–500 active series**, using less than 5% of the free allocation.

Telegraf is the recommended metrics agent for three reasons: its push-based model Best of Web works behind NAT without inbound connectivity, its 300+ input plugins cover every needed metric category, Medium and its ARM64 .deb packages install cleanly. Docker Hub Grafana Alloy (the successor to the now-EOL Grafana Agent) is a strong alternative if you prefer staying purely in the Grafana ecosystem, Medium +3 with a built-in prometheus.exporter.unix component equivalent to node_exporter.

The essential Telegraf input plugins for Firewalla monitoring are:

- **System health**: inputs.cpu, inputs.mem, inputs.disk, inputs.system, inputs.temp (critical for a fanless ARM device)
- **Network interfaces**: inputs.net capturing bytes, packets, errors, and drops per interface (eth0–eth3, br0, WireGuard tunnels)
- **Connection tracking**: inputs.conntrack for ip_conntrack_count vs ip_conntrack_max — the single most important firewall health metric
- **WAN quality**: inputs.ping targeting 8.8.8.8, 1.1.1.1, and your ISP gateway for latency and packet loss; Medium inputs.dns_query measuring resolver response times
- **Security**: inputs.exec running custom scripts to parse Zeek connection counts, blocked flows from /alog/acl-alarm.log, and active device counts from Redis

The Telegraf output pushes directly to Grafana Cloud via HTTP using Prometheus remote_write or InfluxDB line protocol:

```toml
[[outputs.http]]
  url = "https://prometheus-prod-XX.grafana.net/api/prom/push"
  username = "<numeric_user_id>"
  password = "<grafana_cloud_api_key>"
  data_format = "prometheusremotewrite"
  [outputs.http.headers]
    Content-Type = "application/x-protobuf"
    Content-Encoding = "snappy"
    X-Prometheus-Remote-Write-Version = "0.1.0"
```

To survive firmware updates, install Telegraf via a persistence script at `/home/pi/.firewalla/config/post_main.d/install_telegraf.sh` that downloads the ARM64 `.deb`, installs it, copies the config from the persistent config directory, and starts the service. Alternatively, run Telegraf as a Docker container with `--restart always`, which is the approach community members have validated for node_exporter on Firewalla Gold. Firewalla

No Firewalla-specific Grafana dashboards exist in the community. The best starting templates are the **OPNsense Dashboard** project (`github.com/bsmithio/OPNsense-Dashboard`), which uses a similar Telegraf + Grafana architecture GitHub with Suricata panels, and Grafana dashboard **#5955** (Telegraf System Metrics). Both require minor adaptation for Firewalla's interface names and log paths.

Forwarding logs to a cloud-hosted indexable platform

For the log pipeline, the agent and destination choices matter significantly for both cost and capability.

Fluent Bit is the optimal log shipper for the Firewalla's constrained environment. At **~26 MB RAM** with official ARM64 builds, it's purpose-built for edge and IoT devices. Onidel It supports output plugins for Loki, Splunk HEC, Elasticsearch, Axiom, S3, and dozens of others. Grafana Alloy (~50–80 MB) is the runner-up, particularly compelling if you're already using it for metrics. Avoid Fluentd (~120 MB, 80% CPU at 5K events/sec) DEV Community and Vector (~50–100 MB) as unnecessarily heavy for this use case. **Promtail reaches end-of-life in March 2026** Grafana and should not be used for new deployments. Ricsanfre

The practical log collection strategy combines two mechanisms. First, rsyslog (already installed on the device) handles standard syslog forwarding with zero additional overhead. Second, Fluent Bit tails the Zeek JSON logs from `/bspool/manager/` and application logs from `/alog/`, which contain the rich network telemetry that syslog misses. The rsyslog `imfile` module can also tail these files, but Fluent Bit provides better parsing, buffering, and multi-destination output. GitHub

Here is how the cloud log platforms compare for a home user ingesting approximately 1–5 GB/day of firewall logs:

Platform	Free tier	Retention	Query language	Monthly cost at 5 GB/day
Grafana Cloud (Loki)	50 GB/month	14 days	LogQL	$0 (within limit at ~1.7 GB/day)
Axiom	500 GB/month	30 days	APL (Kusto-inspired)	$0
New Relic	100 GB/month	8 days	NRQL	$0 (within limit at ~3.3 GB/day)
Elastic Cloud	14-day trial only	—	KQL/Lucene	~$95/month minimum
Splunk Cloud	None	—	SPL	~$15,000+/year
Datadog	Negligible	—	Proprietary	~$0.10/GB + $1.70/M events

Grafana Cloud Loki is the top recommendation because it bundles with your metrics in a single platform, and LogQL handles firewall log analysis well — filtering by labels (source IP, dest port, log type) and JSON parsing of Zeek structured logs. Grafana At 1.7 GB/day effective free limit, it covers most home networks. X Julius **Axiom is the strongest alternative** with a remarkably generous 500 GB/month free tier, 30-day retention, and a powerful APL query language inspired by Kusto. Hyperping For a user who generates more verbose logs or wants longer retention without paying, Axiom is genuinely superior on free-tier economics.

Splunk and Elastic are non-starters for cost-conscious home use. Splunk Cloud has enterprise-only pricing with no permanent free tier. SaaSworthy Elastic Cloud starts at roughly $95/month. Both offer best-in-class search (SPL and KQL respectively), but the cost is prohibitive for personal network monitoring.

Community projects worth knowing about

The Firewalla monitoring ecosystem is small but functional. The most important resources are:

- **Syslog forwarding gist** (gist.github.com/mbierman/f3d184b65e0f4de6fa75a4a5d5145426) Firewalla Firewalla — The most battle-tested community solution for log export, including an advanced rsyslog configuration using imfile to capture Zeek logs from /bspool/manager/. Multiple community members have contributed improvements over several years. github
- **Official MSP API examples** (github.com/firewalla/msp-api-examples) GitHub Firewalla — Node.js examples for paginated flow export, alarm retrieval, device bandwidth tracking, and Discord alarm forwarding. GitHub The starting point for any API-driven pipeline.
- **go-firewalla** (github.com/marekq/go-firewalla) — A Go-based tool that downloads device details, flow logs, and alarms from the MSP API GitHub to local CSV files. GitHub Directly useful as a data extraction pipeline that could feed into any backend.
- **Node exporter Docker setup** (documented on Firewalla community forums) — A proven single-command Docker deployment of Prometheus node_exporter on Firewalla Gold, providing comprehensive system metrics. The Docker approach cleanly avoids overlay filesystem issues. Firewalla
- **hass-firewalla-ng** (github.com/shanelord01/hass-firewalla-ng) — The most mature Home Assistant integration, which can serve as a bridge to monitoring systems. GitHub Its firewalla.search_flows and firewalla.search_alarms services demonstrate comprehensive MSP API v2 usage. GitHub
- **Firewalla open source repo** (github.com/firewalla/firewalla, 587 stars) GitHub — Essential for understanding internal data structures, Redis key patterns, and log file locations. Reading the source is the only reliable way to map the undocumented internal data model.

Notably absent are any published Firewalla-specific Grafana dashboard JSON files, Prometheus exporters, or Splunk/ELK integration apps. The Firewalla community forum has a Grafana feature request thread open since 2019, with an official response acknowledging it as "one of the features that's not popular, but important" — but no implementation has materialized. Firewalla +2

Recommended architecture for comprehensive cloud-hosted visibility

The cleanest cost-effective architecture for an experienced IT professional looks like this:

```
Firewalla Gold SE (ARM64, 4 GB RAM)
├── Telegraf (Docker, ~30-60 MB RAM)
│   ├── inputs: cpu, mem, disk, net, conntrack, ping, dns_query, temp, exec
│   └── output: Prometheus remote_write → Grafana Cloud Mimir
├── Fluent Bit (Docker, ~26 MB RAM)
│   ├── inputs: tail /bspool/manager/*.log (Zeek JSON), tail /alog/*.log
│   └── output: HTTP → Grafana Cloud Loki (primary), Axiom (optional secondary)
└── rsyslog (native, ~5 MB RAM)
    └── forward *.* → Grafana Cloud Loki syslog endpoint

          ↓ HTTPS (outbound only, no inbound ports)

Grafana Cloud (Free Tier)
├── Mimir: ~150-500 metric series, 14-day retention
├── Loki: ~1-2 GB/day firewall logs, 14-day retention
├── Grafana: dashboards + alerting
└── Cost: $0/month
```

Total RAM overhead on the Firewalla is approximately 90–120 MB (Telegraf + Fluent Bit containers), leaving roughly 3.8 GB for Firewalla's core services — comfortable but worth monitoring. Both containers should use `--restart always` and `--memory 128m` limits as safety guardrails.

For persistence across firmware updates, create `/home/pi/.firewalla/config/post_main.d/monitoring_stack.sh`:

```bash
#!/bin/bash
# Re-pull and start monitoring containers after firmware update
docker pull arm64v8/telegraf:latest
docker pull fluent/fluent-bit:latest-arm64
# Start with configs mounted from persistent storage
docker run -d --restart always --name telegraf --network=host \
  -v /home/pi/.firewalla/config/telegraf.conf:/etc/telegraf/telegraf.conf:ro \
  -v /:/hostfs:ro arm64v8/telegraf:latest
docker run -d --restart always --name fluent-bit --network=host \
  -v /home/pi/.firewalla/config/fluent-bit.conf:/fluent-bit/etc/fluent-bit.conf:ro \
  -v /bspool/manager:/logs/zeek:ro \
  -v /alog:/logs/alog:ro \
  fluent/fluent-bit:latest-arm64
```

Store telegraf.conf and fluent-bit.conf in /home/pi/.firewalla/config/ where they survive updates. Firewalla GitHub If your log volume exceeds Grafana Cloud's free 50 GB/month, X add Axiom as a secondary Fluent Bit output — its **500 GB/month free tier with 30-day retention** provides generous overflow capacity at zero cost. Hyperping

For the MSP API data (per-device bandwidth, categorized flows, security alarms), a lightweight cron-based Python or Go script polling the API every 5 minutes and pushing to Grafana Cloud via Prometheus pushgateway or Loki structured logs completes the picture. The go-firewalla project provides a working starting point. GitHub The MSP subscription costs ~$39/year Firewalla but unlocks the only official programmatic access to application-level firewall data that isn't available through Zeek logs alone. Firewalla

Conclusion

The Firewalla Gold SE is a capable but monitoring-unfriendly appliance compared to pfSense or OPNsense, which offer native Telegraf packages, SNMP agents, and standardized syslog. Firewalla's overlay filesystem, Firewalla undocumented internal data model, and absent export features mean every integration requires SSH-level engineering. Firewalla Firewalla The reward, however, is substantial: Zeek generates exceptionally rich network flow data Zeek Medium that most consumer firewalls never expose. The Telegraf + Fluent Bit + Grafana Cloud architecture delivers real-time dashboards and searchable logs at zero recurring cost, with approximately 90 minutes of initial setup for someone comfortable with Docker and Linux. The key risk is firmware update fragility — test your persistence scripts after every major Firewalla release, and consider pinning to a specific firmware version if stability matters more than new features.